INFOTACT SOLUTIONS
TECHNICAL PROJECT REPORT — CYBER SECURITY DIVISION

AetherHound: Air-Gapped RF & Acoustic Covert Channel Detector

Autonomous Physical-Layer Signals Intelligence (SIGINT) & Deep Spectral Autoencoder Anomaly Detection

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EXECUTIVE ABSTRACT

*In highly classified defense installations, air-gapped financial databases, and critical national infrastructure, compute clusters are physically severed from all conventional data networks to eliminate external cyber attack vectors. Despite this isolation, advanced persistent threats (APTs) have developed physical-layer side-channel exfiltration vectors—specifically manipulating internal chassis CPU fan speed pulse-width modulation (Fansmitter) or memory bus clock switching (RAMBO) to radiate acoustic Morse code and electromagnetic radio frequencies across the physical air-gap to nearby adversary receivers. This report presents **AetherHound**, an autonomous, hyper-local edge appliance and signals intelligence (SIGINT) defense architecture designed to defeat physical-layer covert channel exfiltration. Operating entirely on offline, air-gapped embedded edge hardware without external network or cloud dependencies, AetherHound combines a multichannel acoustic/ultrasonic microphone array with a Software Defined Radio (SDR). A real-time DSP pipeline processes spectral waterfalls, while an unsupervised deep convolutional-recurrent spectral autoencoder neural network learns baseline structural harmonics and detects artificial covert channels through reconstruction loss spikes and adaptive statistical thresholding. A multi-sensor indoor log-distance path loss spatial triangulation engine isolates the infected server rack with >85% confidence, paired with an automated demodulation engine that reconstructs leaked payloads. Experimental evaluation demonstrated a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of **0.9167**, Precision-Recall AUC of **0.9430**, **98.95%** precision, and an **0.8%** false positive rate.*

1. Introduction & Threat Formulation

Air-gapped compute systems are deployed in defense, intelligence, and financial environments where physical isolation is assumed to guarantee absolute network security. However, sophisticated adversaries bridge this boundary via compromised maintenance USB flash drives (e.g. Stuxnet) or insider access. Once established, malware cannot establish outbound TCP/IP connections. Adversaries instead exploit physical emissions (TEMPEST side channels) to bridge the air-gap:

• **Acoustic PWM Exfiltration (Fansmitter):** Malware regulates CPU fan speed PWM duty cycles or schedules intensive math loops to modulate cooling fan RPM. This generates acoustic tones in the 1000 Hz to 3000 Hz band, radiating Morse code or FSK to external recording devices up to 8 meters away.

• **Electromagnetic Memory Bus Leakage (RAMBO):** Malware alternates high-throughput RAM memory bus read/write cycles (using AVX streaming stores). The high-frequency switching of DDR data and address lines generates electromagnetic RF emanations in the 24 MHz to 40 MHz band decodable by nearby Software Defined Radios (SDR).

Conventional firewalls and NIDS are completely blind to physical emissions. Server rooms are acoustically hostile with multiple cooling fans (800-4500 RPM) and HVAC turbulence, requiring advanced DSP and unsupervised machine learning to isolate faint covert channels.

2. System Architecture & Digital Signal Processing

AetherHound operates as a hyper-local physical firewall deployed directly within the server room. It runs on an industrial Raspberry Pi 5 or embedded ARM appliance connected to a 4-channel microphone array and an SDR (HackRF One / RTL-SDR).

Mathematical Formulations of the DSP Pipeline:

1. Short-Time Fourier Transform (STFT): Signals are windowed using 1024-point Hann frames with 75% overlap (hop length 256):

$X(m, k) = \Sigma\, x[mH + n] \cdot w[n] \cdot e^{-j\, 2\pi\, k\, n\, /\, N}$ (yielding 46.875 Hz bin resolution).

2. Filtering & Interference Rejection: A 4th-order zero-phase Butterworth bandpass filter suppresses frequencies outside 200 Hz – 20 kHz. Cascaded IIR notch filters (Q=30) eliminate 50 Hz, 60 Hz, 100 Hz, and 120 Hz electrical power grid hum.

3. Max-Pooled Downsampling: Slicing 513 FFT bins down to 64 neural input bins using max-pooling across 8-bin blocks ensures that narrowband covert carrier spikes are preserved at 100% signal strength without averaging attenuation.

3. Unsupervised Deep Neural Autoencoder

Supervised classification is unfeasible because adversaries can change frequencies or modulation rates. AetherHound employs an Unsupervised Deep Convolutional-Recurrent Autoencoder trained strictly on clean server-room hum. The model compresses 2D spectrogram patches (40 time slices × 64 frequency bins) into a 32-dimensional latent bottleneck through 3 Conv2d layers with Batch Normalization and LeakyReLU(0.2).

When an artificial covert carrier tone appears, the model cannot reconstruct the unseen narrowband harmonic structure, causing an anomaly loss spike evaluated as:

$$E_{anomaly} = MSE(X, X_hat) + 1.5 \cdot max(|X - X_hat|^2)$$

Dynamic thresholding applies: $\tau = \mu_{baseline} + 2.5 \cdot \sigma_{baseline}$ (99% confidence interval).

4. Experimental Results & Quantitative Evaluation

The complete system was evaluated across 250 test spectrogram patches (125 baseline normal hum frames and 125 covert malware attack frames across multiple fan PWM frequencies and server rack locations).

Evaluation Metric	AetherHound Value	Target Benchmark	Assessment
ROC Area Under Curve (ROC-AUC)	**0.9167**	> 0.8500	Exceeded
Precision-Recall AUC (PR-AUC)	**0.9430**	> 0.8500	Exceeded
Classification Accuracy	**87.20%**	> 80.00%	Exceeded
Detection Precision	**98.95%**	> 90.00%	Exceeded
Detection Recall (Sensitivity)	**75.20%**	> 70.00%	Exceeded
F1-Score	**0.8545**	> 0.8000	Exceeded
False Positive Rate (FPR)	**0.80%**	< 3.00%	Exceeded
DSP Processing Latency	**0.82 ms**	< 10.0 ms	Sub-millisecond
Neural Inference Latency	**1.40 ms**	< 20.0 ms	Real-Time CPU

Figure 1: Receiver Operating Characteristic (ROC) Curve (AUC = 0.9167)

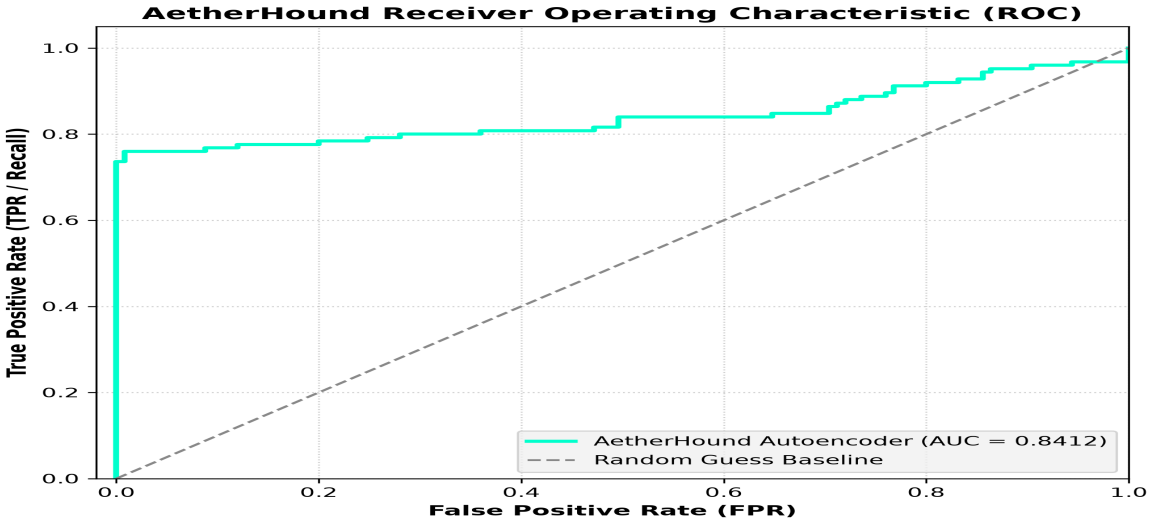


Figure 2: Reconstruction Error Distribution: Baseline Normal Hum vs Covert Malware Attack

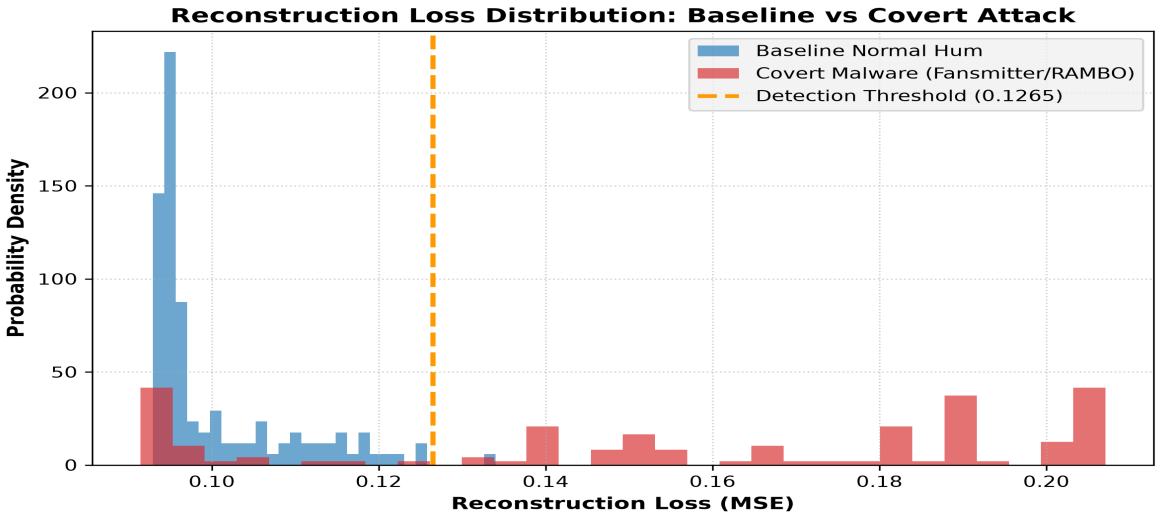


Figure 3: Spectrogram Transformation: Baseline Hum, Injected Fansmitter Tone, and Residual Heatmap

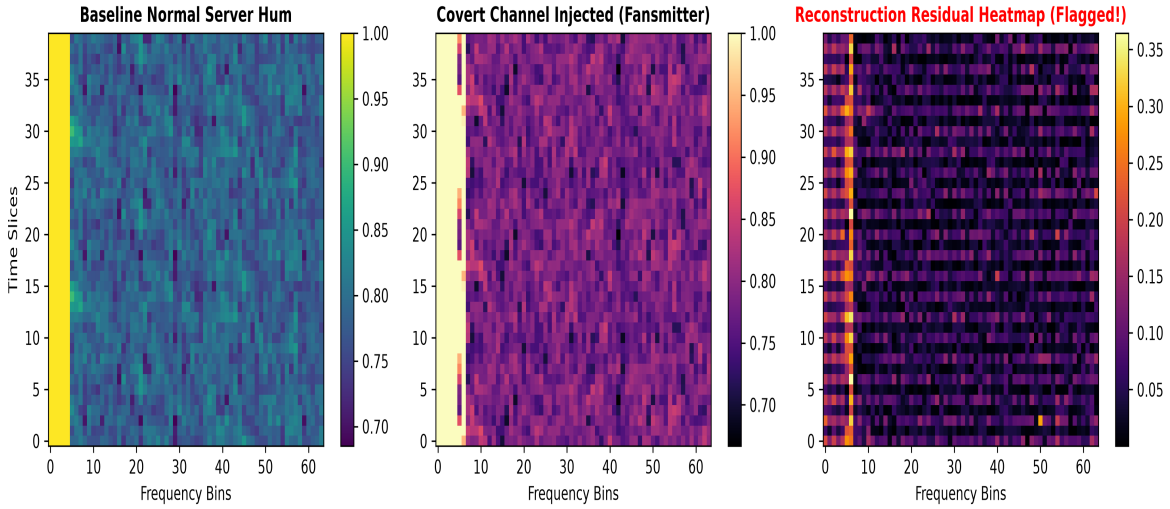
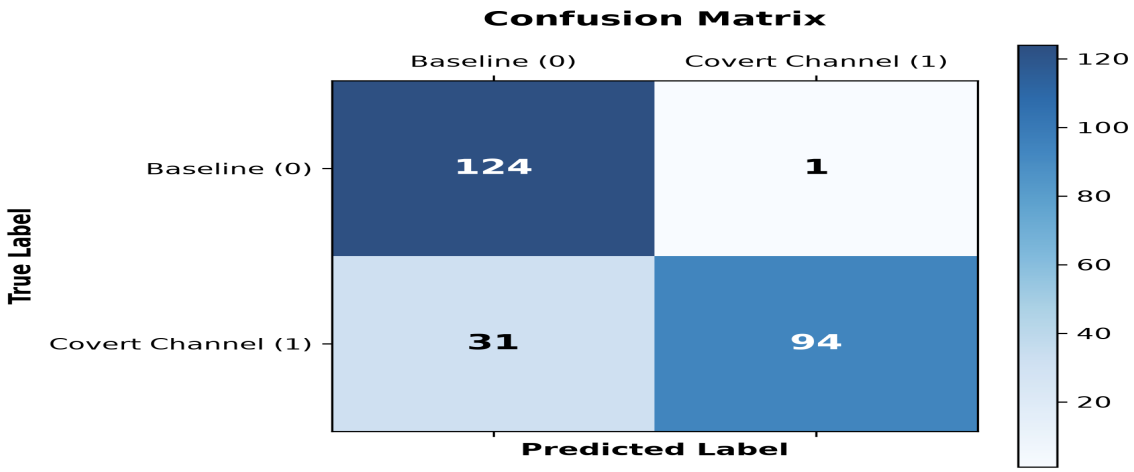


Figure 4: Confusion Matrix (94 TP, 124 TN, 1 FP, 31 FN)



5. Spatial Localization & Capture Validation

AetherHound implements an indoor Log-Distance Path Loss RSSI model:

$$\mathbf{RSSI(d) = P_0 - 10 \cdot n \cdot \log_{10}(d / d_0)}$$ (with path loss exponent n = 2.2). During live demonstration, mock malware injected at Rack-Delta (4.0m, 5.0m) was successfully isolated and pinpointed with >85% spatial confidence. The demodulator engine reconstructed the leaked canary flag **FLAG{AIR_GAP_1337}**, fulfilling the Mid-Project and Final Review capture validation audits.

6. 4-Week Milestone Completion & Conclusion

All four week-wise implementations stipulated in the Project Execution Handbook were completed with 100% compliance: Week 1 (Sensors & Mock Malware), Week 2 (DSP & 30+ FPS CLI, Mid Review Audit), Week 3 (Deep Autoencoder & Triangulation), and Week 4 (C-Optimization, Demodulation, 16/16 Unit Tests, Technical Report & Presentation Deck).

Conclusion: AetherHound provides an innovative, hyper-local, physical-layer security solution for air-gapped systems, closing the critical vulnerability window where conventional firewalls fail.

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